Tutorial 2

Exercise 1

Consider the single-output technology that is associated with the following production function:

$$f(z_1, z_2) = z_1^{\alpha} z_2^{\beta}, \qquad \alpha, \beta > 0.$$

(a) Derive the conditional factor demand function $z(w_1, w_2, q)$.

(b) Derive the cost function $c(w_1, w_2, q)$.

(c) Derive the marginal cost of production.

(d) Derive the supply function $y(w_1, w_2, p)$.

(e) Derive the profit function $\pi(w_1, w_2, p)$.

Exercise 2

A firm uses two inputs, energy (E) and all other inputs (O), with input prices $w = (w_E, w_O) \gg 0$. Its cost function is of the CES form

$$c(w, q) = q\,\Phi(w)^{\frac{1}{1-\sigma}}, \qquad \Phi(w) = a_E\, w_E^{1-\sigma} + a_O\, w_O^{1-\sigma},$$

with $a_E, a_O > 0$ and $\sigma > 0$, $\sigma \neq 1$ (the elasticity of substitution between the two inputs). Write $s_i(w) = w_i z_i(w, q)/c(w, q)$ for the cost share of input i, where $z(w, q)$ is the conditional factor demand.

Shephard's lemma for the firm. If the conditional factor demand $z(w, q)$ is single-valued and $c(\cdot, q)$ is differentiable at w, then

$$\frac{\partial c(w, q)}{\partial w_i} = z_i(w, q) \qquad \text{for each input } i.$$

This is the firm-side counterpart of $\nabla_p e(p, u) = h(p, u)$ for the expenditure function: cost minimisation is the EMP with output q in place of utility u, so the same argument applies.

(a) Show that c is homogeneous of degree one in w and that average cost equals marginal cost and does not depend on q. What does this tell you about returns to scale?

(b) Use Shephard's lemma to derive $z_E(w, q)$ and $z_O(w, q)$. Show that $s_i(w) = a_i w_i^{1-\sigma}/\Phi(w)$ and that z is homogeneous of degree zero in w.

(c) Show that

$$\frac{\partial \ln z_E}{\partial \ln w_E} = -\sigma\,(1 - s_E), \qquad \frac{\partial \ln z_E}{\partial \ln w_O} = \sigma\, s_O.$$

Why must these two elasticities sum to zero?

(d) Input prices change from w^0 to w^1, with $w_i^1 = (1+g_i)\,w_i^0$. Let $s_i^0 = s_i(w^0)$. Show that the exact change in cost at fixed output is

$$\frac{c(w^1,q)}{c(w^0,q)} = \Big[s_E^0(1+g_E)^{1-\sigma} + s_O^0(1+g_O)^{1-\sigma}\Big]^{\frac{1}{1-\sigma}},$$

and that energy use per unit of output changes by the factor $(1+g_E)^{-\sigma}\big[c(w^1,q)/c(w^0,q)\big]^{\sigma}$.

(e) Take an energy-intensive chemicals plant in 2022: $s_E^0 = 0.30$, $\sigma = 0.4$, $g_E = 93\%$ and $g_O = 5.9\%$. Compute

(i) the increase in cost if the firm kept its input mix unchanged, $w^1 \cdot z(w^0,q)/c(w^0,q) - 1$;

(ii) the exact increase in cost, using (d);

(iii) the change in energy use per unit of output, and the new energy cost share s_E^1.

Explain why (i) must be at least as large as (ii) for *any* technology, not only CES.

(f) Now suppose the plant is one of many identical price-taking firms with this technology, and industry demand has constant price elasticity $\varepsilon_D = -1$. Compute the change in the output price, in industry output and in total energy use. In 2022, EU industry cut natural-gas use by 15% while industrial output rose by about 5%. Which part of the model's prediction is in line with this, and which is not? What is the most likely reason for the gap?

Exercise 3

A firm is observed T times. In observation $t = 1,\dots,T$ it faces prices $p^t \gg 0$ and chooses the production vector (netput) $y^t \in \mathbb{R}^L$, with positive entries for outputs and negative entries for inputs. We say that a production set $Y \subseteq \mathbb{R}^L$ *rationalises* the data $\{(p^t,y^t)\}_{t=1}^T$ if, for every t, $y^t \in Y$ and $p^t \cdot y^t \geq p^t \cdot y$ for all $y \in Y$. The data satisfy the *Weak Axiom of Profit Maximisation* (WAPM) if

$$p^t \cdot y^t \;\geq\; p^t \cdot y^s \qquad \text{for all } s,t \in \{1,\dots,T\}.$$

The lectures show that WAPM is necessary: if some production set rationalises the data, the data satisfy WAPM. This exercise shows that it is also sufficient.

(a) Suppose the data satisfy WAPM, and define

$$Y^O = \big\{y \in \mathbb{R}^L : p^t \cdot y \leq p^t \cdot y^t \text{ for all } t = 1,\dots,T\big\}.$$

Prove that Y^O rationalises the data.

(b) Prove that Y^O is

(i) *closed*: whenever $y^n \in Y^O$ for all n and $y^n \to y$, then $y \in Y^O$;

(ii) *convex*: $y, y' \in Y^O$ and $\lambda \in [0,1]$ imply $\lambda y + (1-\lambda)y' \in Y^O$;

(iii) satisfies *free disposal*: $y \in Y^O$ and $y' \leq y$ imply $y' \in Y^O$.

Combining (a) and (b) with the necessity result from the lectures, state what you have shown. What does it imply about whether convexity of the production set can be tested with finite data?

(c) Prove that every production set that rationalises the data is a subset of Y^O.